Protecting the Open Internet

We Chose Encryption

Identity Needs Us

Real World Crypto · Taipei · March 2026

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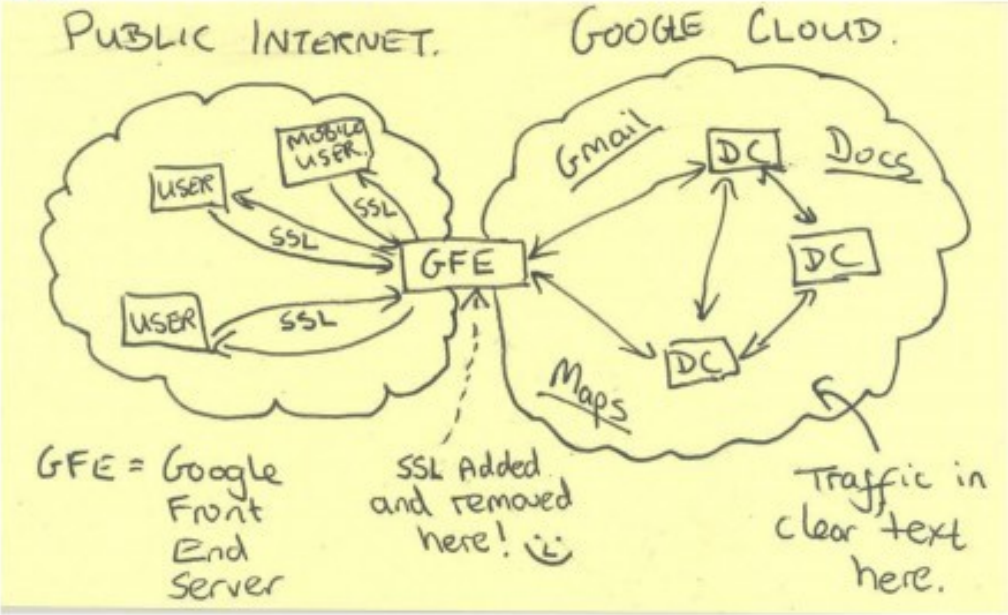
Encryption Everywhere

How we got here, and what it took

TOP SECRET//SI//NOFORN

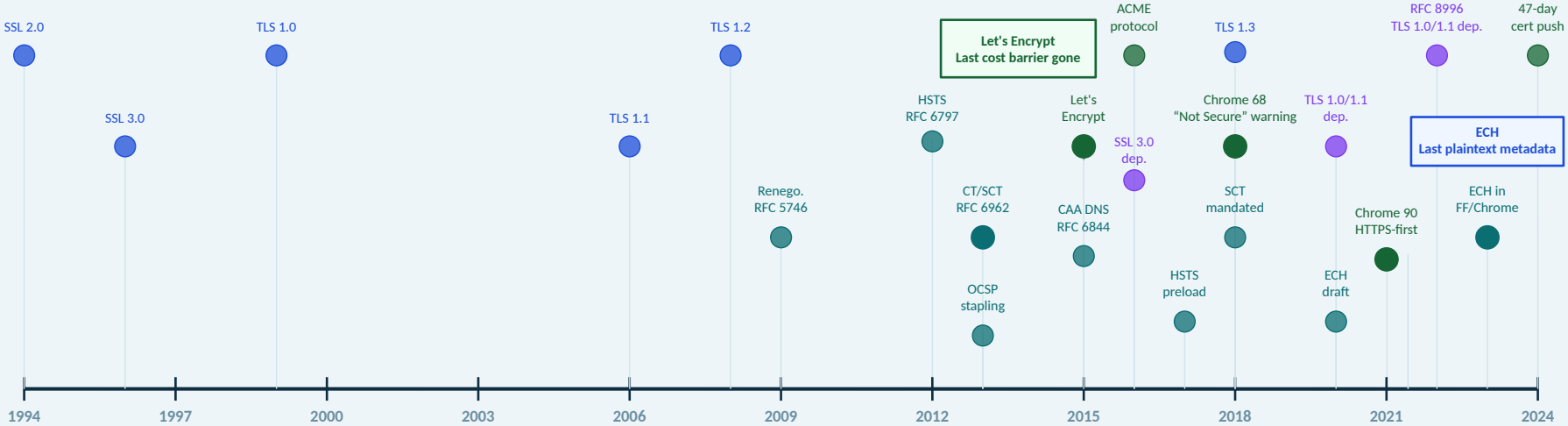
Current Efforts - Google

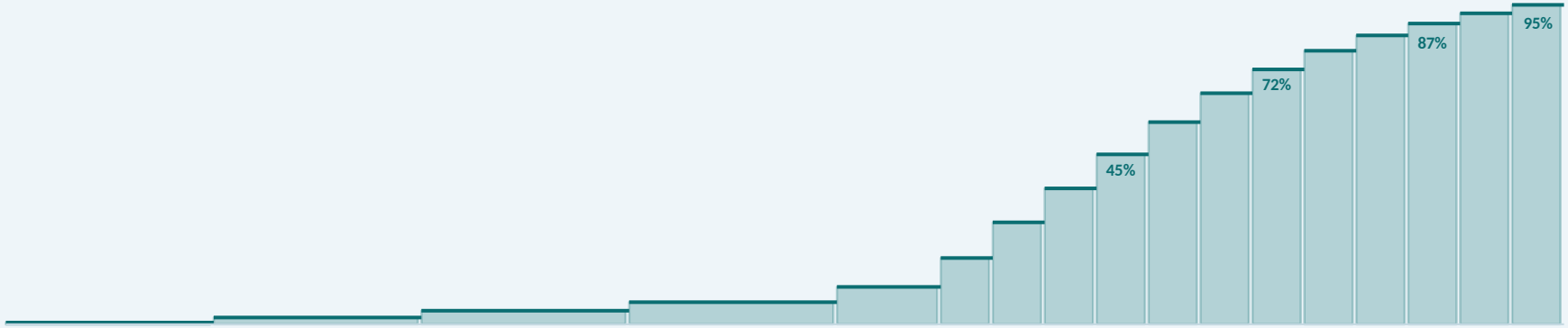


TOP SECRET//SI//NOFORN

NSA / GCHQ MUSCULAR program · Washington Post, October 30, 2013 · Snowden documents

Encrypting the Internet · SSL/TLS and the ecosystem around it





HTTPS % of Chrome page loads (Google Transparency Report)

Protocol versions · Hardening / enhancements · Infrastructure & scale · Deprecations (victories)

5

Major tracks

- **Standards & Protocols**
- **Hardening against whole vulnerability classes**
- **Infrastructure removing barriers**
- **Ecosystem pressure**
- **Policy mandates**

All five. At once.

FIDO / Passkeys

What adoption has actually looked like

0ktapus · 2022 · Targeted 130+ orgs

Twilio

SMS phishing · August 2022

Employees received targeted SMS

Clicked convincing fake Okta login

Entered credentials ✓

Entered TOTP code ✓

Attacker relayed TOTP in real time

Attacker logged in.

209 customers compromised.

v/s

Cloudflare

Same attacker · Same campaign

Employees received same SMS

3 employees clicked, entered credentials

Entered credentials ✓

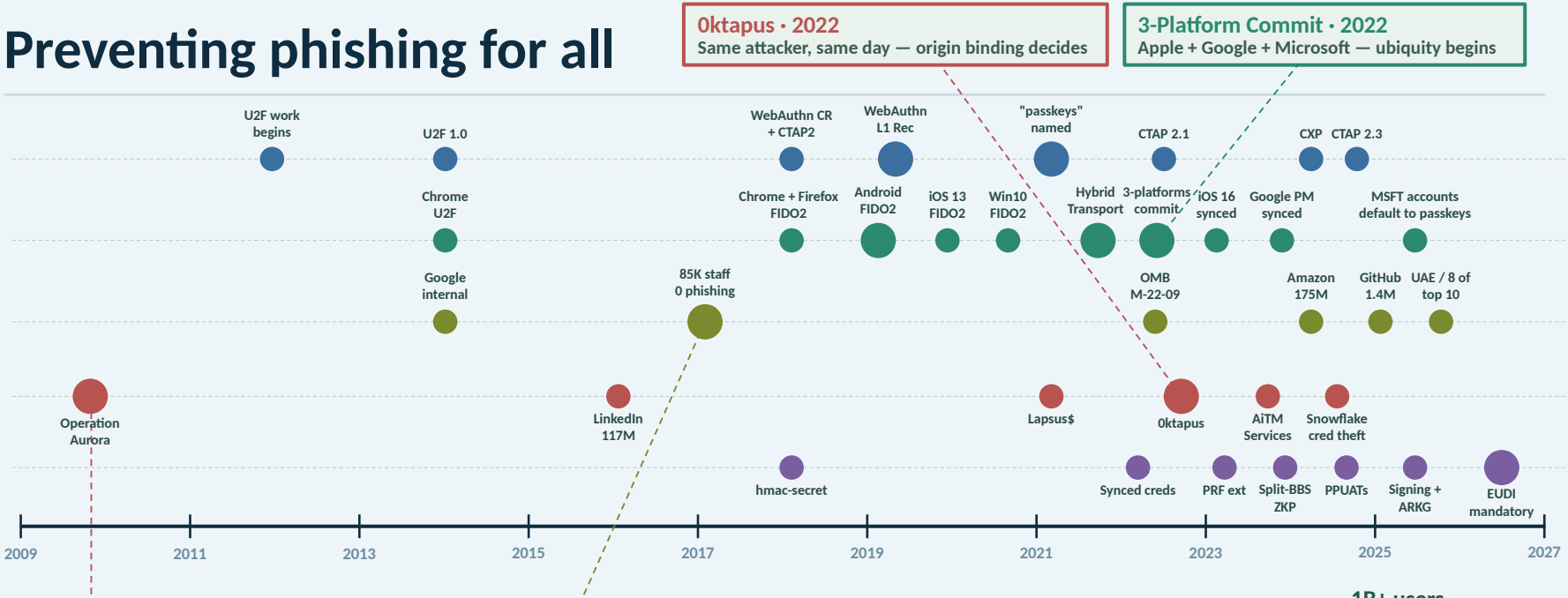
Attacker tried to log in 🚫

FIDO origin binding: credential cryptographically tied to real domain

Phishing site gets nothing.

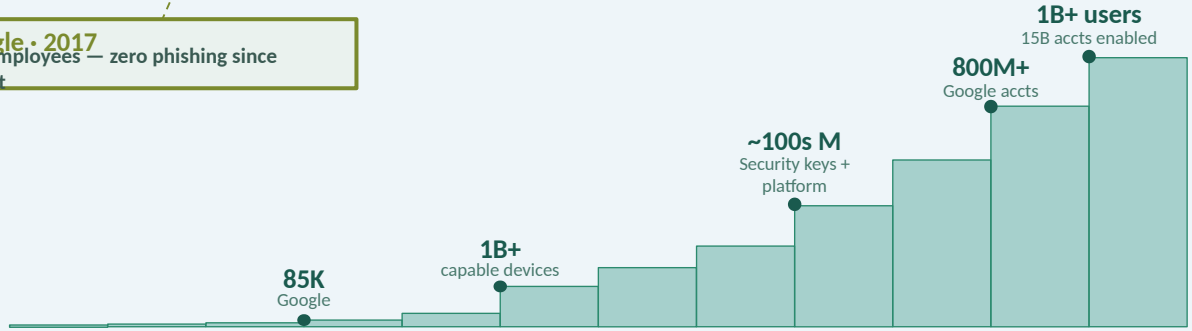
Same attacker, same day; one difference: origin-bound credentials vs. phish-able codes or approvals

Preventing phishing for all



Operation Aurora · 2009
Drives Yubico + Google to create U2F

Google · 2017
85K employees — zero phishing since rollout



Passkey-enabled accounts & users (mixed metrics across vendors — shape is directional, not precise)

Spec / standards · Platform · Deployment · Incidents · Capability

The Competition Was Never Other Strong Auth

Security teams optimized for

- Phishing resistance
- Hardware binding
- Attestation (who made this authenticator)
- No shared secrets
- Synced or hardware: both are passkeys

Product teams actually faced

- Username + password
- "Change every 90 days"
- **SMS OTP**
- **Password reuse across 40 sites**
- **No MFA at all (most accounts)**

Good Specification. Enormous Implementation Surface Area

Attestation

Who made this key? The spec answered it. What to do with the answer? Competing philosophies that still have not fully resolved.

Naming

"Resident key" became "discoverable credential" mid-deployment. Real applications had already shipped the old term.

UV vs UP

User Verification vs User Presence. Two letters. Genuine security regressions at real companies.

RP onboarding

Getting 10,000 relying parties to implement correctly isn’t a cryptography problem. It is a developer relations problem.

Synced passkeys

Hardware advocates did not want them. Platform teams did not want hardware dependency. Both shipped anyway.

Synced Credentials Changed the Goalposts

We traded a recovery problem for a key management problem.

Device-bound credentials

- Passkeys never leave device
- Max security for security-conscious users
- Adoption: mostly high risk or at risk environments and cohorts
- Risk: lose all your devices, lose access

Synced credentials

- Passkeys in iCloud / Google Password Manager
- Cloud provider now in the trust chain
- Adoption: anyone with a modern device

- Risks: recovery often possible, but may be phishable. Key protection varies

Copyable credentials move the key management problem to teams with thousands of security engineers.

Passkeys Are Used For Encryption Now

PRF/hmac-secret: deterministic secret from a salt, scoped to credential + origin. AuthN + encryption

wwWallet	Open-source identity wallet. Encryption keys derived from passkey — wallet operator never holds plaintext credentials.
WhatsApp	End-to-end encrypted backup keys derived from passkey.
Dashlane	Vault encryption entirely from passkey PRF — key exists nowhere except the credential.
AGE (age-plugin-fido2-hmac)	File encryption with a security key or platform authenticator.
Bitwarden	Vault key derivation from passkey PRF.

It Needed More Than Math

Standardization	Implementation	UX	Usability / Accessibility
Performance	i18n	Policy	Certification

14+ years and passkeys are still not done.

All of this is needed for digital identity, but with less runway.

Digital Identity

The train left the station. Which track is it on?

Keypairs vs. Attributes — Different Problem

Keypairs

Prove you control a private key.
Cryptographic, binary, hardware-bindable.
FIDO does this at scale.

Attributes / Predicate proofs

Prove something about who you are.
Age >= 18. Licensed to drive. Citizen. Resident.
Requires an issuer. Requires selective disclosure.
Should not require building a surveillance mechanism.

The Harm Being Done Right Now

This is the current baseline, not a future risk.

- Age verification laws require everyone including children to obtain and upload full government ID and biometrics to third-party services. Data is often retained, correlated, reused.
- Every compliance attempt creates a new identity data silo. None of them talk to each other. All of them are likely to get breached eventually.
- This is what we are protecting against: the low-friction, well-intentioned, surveillance-by-default present.

The Wallet Transition Is a Fork

Wallets are being mandated. The question is which kind.

Without ZKP

Every issuance can be logged
Every presentation is linkable
Issuer + verifier collab can determine:
- what you proved
- to whom
- when

Convenient. Functional.
Very difficult to move away from

With ZKP

Selective disclosure
Unlinkable presentations
Verifier learns only what you showed
- not your identity
- not other attributes
- not that you have presented before

Everlasting privacy if designed right.

The Attractor State

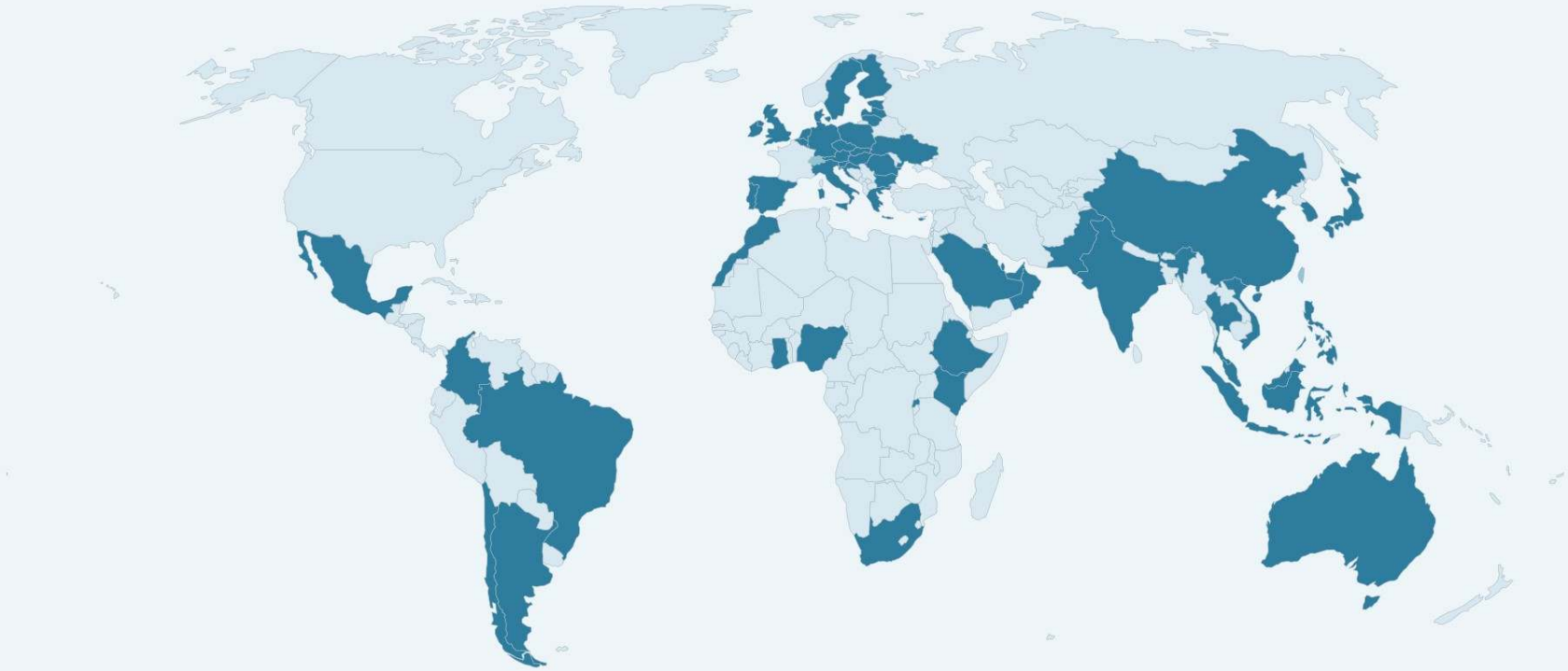
"I live in Sweden. I use BankID every day.
It is the most convenient identity tool I have ever touched.
It also has zero privacy.
Every verification is logged, linked, and attributable.
Nobody complains — because it just works."

10M+	100+	~0	High
Swedish users	banks & services	privacy properties	adoption rate

It's low-friction, functional, zero privacy, and no obvious path back.

Digital Identity Schemes

62 jurisdictions with enacted or announced national digital identity schemes



Digital identity enacted

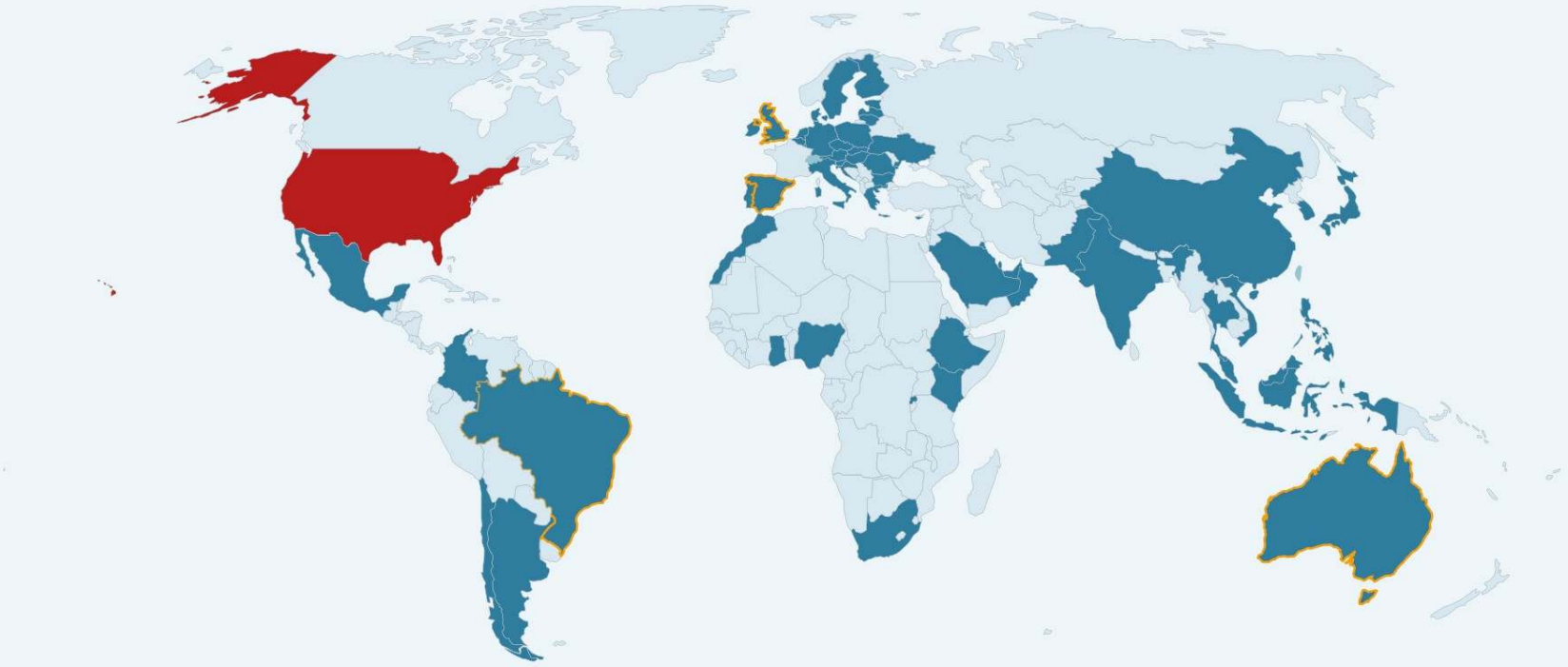
Announced / launching 2026

62 JURISDICTIONS

Rough data!

Age Verification Mandates

Overlaid: jurisdictions with enacted or imminent age verification / children's online safety laws



Digital ID enacted | Announced 2026 | AV enforcing (no nat'l ID) | AV imminent

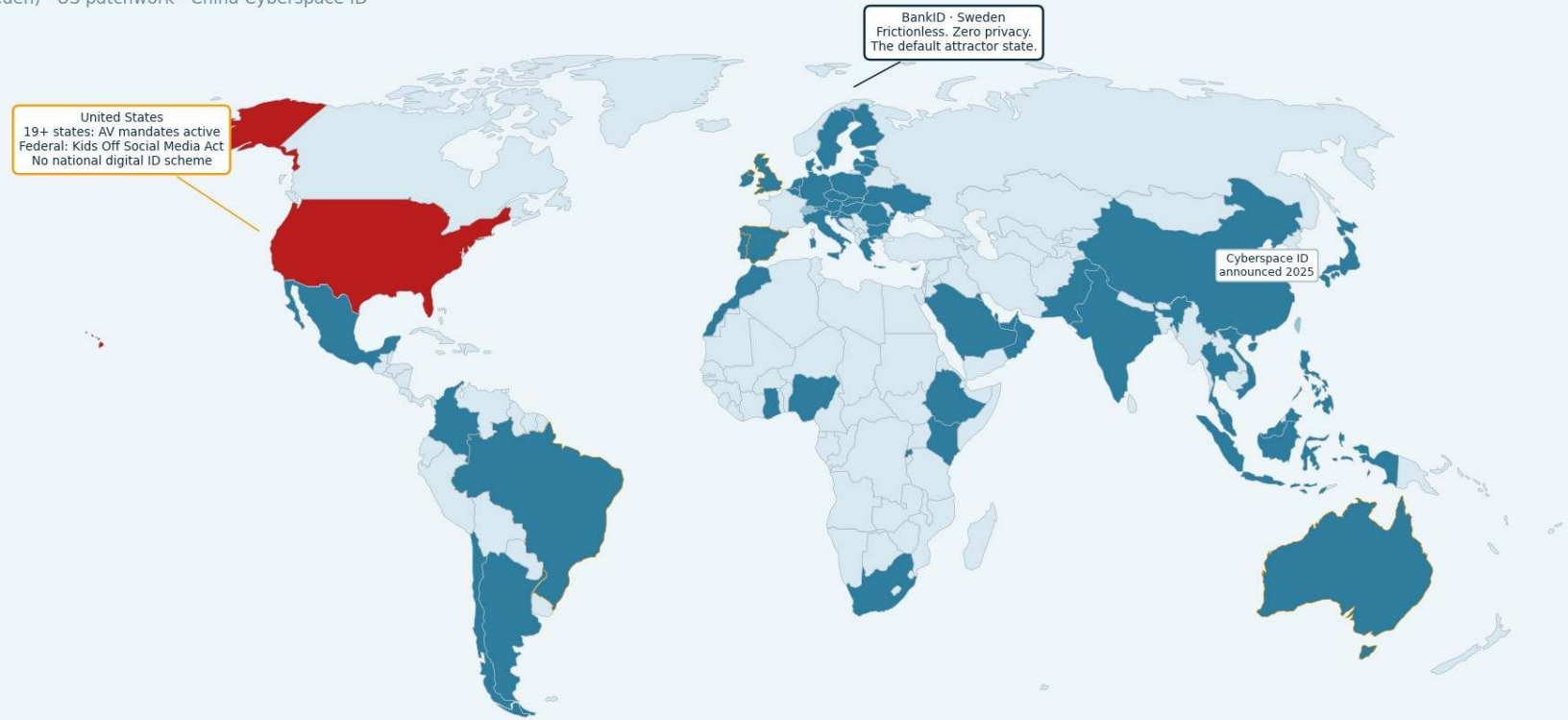
Gold border = Digital ID + AV enforcing

62 ID · 7+ AV LAWS

Rough data!

Key Reference Points

BankID (Sweden) · US patchwork · China Cyberspace ID



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62 ID · 7+ AV LAWS

Rough data!

The Legislative Sweep

55+

jurisdictions with digital identity enacted or announced

EU 27	EUDI Wallet · mandatory by December 2026
India	Aadhaar · 1.38 billion enrolled
Brazil	Gov.br · 171 million users
Vietnam / Thailand	VNeID 62M+, ThaID 40M+
Switzerland	Swiyu · passed referendum Sep 2025, launching 2026
United Kingdom	GOV.UK wallet, digital driving licence June 2025
UAE / Saudi Arabia	UAE Pass (12k+ services), Nafath mandatory 2024
South Korea / Japan	Mobile national ID, 2024-25 rollouts
China	Cyberspace ID · announced 2025, Belt-and-Road risk
Australia	Digital ID Act 2024 / myID

Age Verification · enforcing now:

UK (July 2025) · Australia (Dec 2025) · France (June 2025) · Brazil (March 2026) · 19+ US states

France AV decree notable: explicitly requires anonymity layers. This is the closest any legislature has come to mandating ZKP by design.

The Cryptography Problem

Nobody is post-quantum safe yet

Select ZKP Anonymous Credential Schemes

Scheme	Org	Existing HW	Everlasting Privacy	Proof Status	Notes
Longfellow	Google	✓	✓*	~ ROM*	Ligero interactive Fiat-Shamir
Crescent	Microsoft	✓	✗	✓ Proven	Groth16/BN254 **computational ZK**
BBS#	Orange	✓	✓	⚠ Partial	Novel composition is under review
Vega	MS Research	✓	✓	✗ None	135x faster precomp · unaudited
Split-BBS	Yubico/SPRIND	✗	✓	⚠ Partial	BLS12-381 · HW prototype only Abandoned for now

All 5 schemes vulnerable to credential forgery post-Q-Day. Crescent is the only scheme without everlasting privacy.

The Post-Quantum Gap

ALL vulnerable Credential forgery post-Q-Day	ECDSA/RSA/BLS all broken by Shor's. Every issued credential forgeable.
4 of 5 schemes Everlasting privacy (ZK hiding)	Information-theoretic hiding. Transcripts cannot be retroactively de-anonymized.
4 of 5 schemes Existing secure hardware support	P-256/ECDSA native in Secure Enclave, StrongBox, TPM 2.0, production smartcards.
PQC-safe credentials with ZK presentations None deployed	ML-DSA in ZK circuits: ~100KB transcripts, ad-hoc assumptions, verification too slow. Research exists (CCS 2024, ePrint 2024/131). Gap from paper to deployment is wide.

"If we wait for mass adoption before solving post-quantum anonymous credentials, ACs risk being dead on arrival." — Cloudflare Research

Call to Action

Three asks. One window.

It Needs More Than Math

True for TLS. True for FIDO. True for ZKP anonymous credentials.

Standardization	**Implementation**	**UX**	**Usability / Accessibility**
Performance	**i18n**	**Policy**	**Certification**

Even when the math is right, everything else is where deployments die

Right now: ZKP math for PQC credentials isn’t finished

Three Asks

01

Join IETF ZIP

Zero-knowledge Identity Proofs mailing list

The list is new and the floor is open.

02

Bring your ZKP scheme to wwWallet / SIROS / Yubico

Apache 2.0 · open source · web-native · GUnet / SUNET / Yubico / SIROS Foundation / ISRG / others

Already implementing Longfellow, uses passkeys w/ encryption, testing ZKP w/ actual hardware.
YubiKey 5.8 preview: hardware-bound passkey-based digital signatures with ARKG

03

Solve PQC-safe ZK credential signatures

This room is the right room

The research exists in pieces. The deployed scheme does not.

We Chose Encryption.

Identity Needs Us.

Those building a worse version are well-funded and moving fast.
The window is still open, but it won't stay that way.

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IETF ZIP
https://mailarchive.ietf.org/arch/browse/zip/

SIROS / wwWallet:
https://siros.org/

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